

Group 2

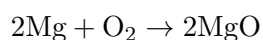
A Level Chemistry

The Group 2 metals

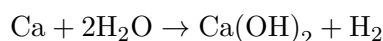
Group 族 2 holds the metals magnesium, calcium, strontium and barium. They all have two outer electrons, which they lose to form 2+ ions. Going **down** the group, the atoms get larger and the outer electrons are easier to lose, so the metals get **more reactive** —their **reactivity** 反应活性 increases down the group.

Reactions of the elements

With **oxygen**, they burn to form an oxide:



With **water**, they form a hydroxide and hydrogen. The reaction gets faster down the group:



Magnesium is slow with cold water but reacts fast with steam, giving MgO and H₂.

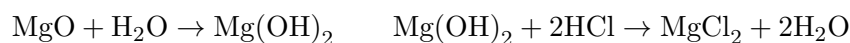
With **dilute hydrochloric or sulfuric acid**, they form a salt and hydrogen:



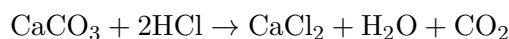
With sulfuric acid the reaction slows down the group, because the **sulfates** 硫酸盐 formed (such as BaSO₄) are insoluble and coat the metal.

Reactions of the compounds

The **oxides** 氧化物 and **hydroxides** 氢氧化物 are basic. They react with water and with dilute acids:



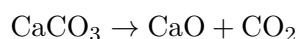
The **carbonates** 碳酸盐 react with dilute acids to give a salt, water and carbon dioxide:



Thermal decomposition

Thermal decomposition 热分解 means breaking a compound apart by heating it.

- the carbonates break into the oxide and carbon dioxide:



- the **nitrates** 硝酸盐 break into the oxide, brown nitrogen dioxide gas and oxygen:



The **thermal stability** 热稳定性 of both the carbonates and the nitrates **increases** down the group. A larger metal ion pulls less on the carbonate or nitrate ion, so the compound is harder to break apart—it needs a higher temperature. So magnesium carbonate decomposes most easily, and barium carbonate is the hardest.

Trends in solubility

Compound	Trend in solubility 溶解度 down the group
hydroxides	increase ($\text{Mg}(\text{OH})_2$ almost insoluble; $\text{Ba}(\text{OH})_2$ soluble)
sulfates	decrease (MgSO_4 soluble; BaSO_4 insoluble)

From these trends you can predict the properties of the next element down, radium, and of its compounds.